

THE NONASSOCIATIVE HINDMAN AND ELLIS CONJECTURES ARE FALSE

LITERATURE-STATUS PACKET FOR ARXIV:1106.4735

1. THE TWO SOURCE CONJECTURES

Let T be the free binary system on one generator, let T_m be its terms with m leaves, and let A be the union of the finite-dimensional probability simplices supported on the T_m . Moore's Conjecture 1.3 proposes a nonassociative Hindman theorem for every coloring $c : T \rightarrow [0, 1]$. Conjecture 1.4 proposes that every compact convex subsystem of the finitely additive probability measures on an arbitrary binary system contains an idempotent. Both appear together on official PDF page 4:

Conjecture 1.3. If $c : \mathbb{T} \rightarrow [0, 1]$ and $\epsilon > 0$, then there is an $r \in [0, 1]$ and an increasing sequence μ_i ($i < \infty$) of elements of $\mathbb{A}$ such that whenever t is in $\mathbb{T}_m$ and $i_0 < \dots < i_{m-1}$ is admissible for t

$$|c(t(\mu_{i_0}, \dots, \mu_{i_{m-1}})) - r| < \epsilon$$

Admissibility is a technical condition which will be defined later. For now it is sufficient to mention two of its properties. First, if $m - 1 \leq i_0 < \dots < i_{m-1}$, then $i_0 < \dots < i_{m-1}$ is admissible for any element of $\mathbb{T}_m$. Additionally any increasing sequence of m integers is admissible for some element of $\mathbb{T}_m$. In particular, if $c(t)$ is required to depend only on $\#(t)$, then the above conjecture reduces to Hindman's Theorem.

It was shown in [17] that even a weak form of Hindman's theorem for $(\mathbb{A}, \wedge)$ is sufficient to prove that Thompson's group F is amenable. The relationship between F 's amenability and the Ramsey theory of $(\mathbb{A}, \wedge)$ becomes even more apparent when one attempts to generalize Ellis's Lemma. The extension of a binary operation $\star$ on a set S mentioned above can be generalized so as to extend $\star$ to the space $\ell^\infty(S)^*$, which contains the set $\text{Pr}(S)$ of all finitely additive probability measures on S :

$$\mu \star \nu(f) = \int \int f(x \star y) d\nu(y) d\mu(x)$$

This leads to the following conjectural extension of Ellis's Lemma.

Conjecture 1.4. If $(S, \star)$ is a binary system and $C \subseteq \text{Pr}(S)$ is a compact convex subsystem, then there is a μ in C such that $\mu \star \mu = \mu$.

The following result provides an intriguing strategy for proving F 's amenability.

2. THE LATER LITERATURE ANSWER

Moore's later paper arXiv:1807.05469 states explicitly that free binary systems do not support idempotent means and that both source conjectures are refuted:

the amenability of a specific group. I previously (and incorrectly) claimed to have proved the existence of an idempotent mean on a free binary system [6]. In this article I will prove that free binary systems does not support idempotent means, refuting Conjecture 1.4 of [5]. I will also refute Conjecture 1.3 of [5], which was an extension of Hindman's theorem to nonassociative binary systems. The question of

This is an exact ask/answer match, not merely an obstruction to one proposed proof. A free binary system is a binary system, so failure there is a counterexample to the universal quantifier in Conjecture 1.4. The same free system on one generator is the object of Conjecture 1.3.

3. COUNTEREXAMPLE MECHANISM

For completeness, here is the core of the later proof. If $(S, *)$ is freely generated, let $\#(s)$ be the number of generators in the unique term for s , write $S_n = \{s : \#(s) = n\}$, and recursively define

$$T_0 = S, \quad T_{p+1} = (S \setminus Z) * T_p, \quad s \in Z \iff s = a * b \text{ with } b \in T_{\#(a)}.$$

Suppose that μ were idempotent and put $r = \mu(Z)$. Freeness and finite additivity give $\mu(S_n) = 0$ and $\mu(T_n) = (1 - r)^n$ for every n . If $r > 0$, choose n with $(1 - r)^n < r$; decomposing Z by the left size forces $r \leq (1 - r)^n$, a contradiction. If $r = 0$, every section $\{t : s * t \in Z\} = T_{\#(s)}$ has μ -measure one, so $\mu * \mu(Z) = 1 \neq 0 = \mu(Z)$. The later paper records the conclusion and then turns to the second conjecture:

If $r = 0$, then for every $s \in S$,

$$\mu(\{t \in S \mid s * t \in Z\}) = \mu(T_{\#(s)}) = (1 - r)^{\#(s)} = 1.$$

However by Fact 2, $\mu(Z) = \mu * \mu(Z) = 1 \neq 0 = \mu(Z)$, which is also a contradiction. This refutes Conjecture 1.4 of [5] which asserts that whenever $(S, *)$ is a binary system, every compact convex subsystem of $M(S)$ contains an idempotent.

3. THE NONASSOCIATIVE ANALOG OF HINDMAN'S THEOREM IS FALSE

In this section we will refute Conjecture 1.3 of [5], which was suggested as a nonassociative analog of Hindman's Theorem. For the duration of the section, we will let $(S, *)$ denote the free binary system on one generator **1**. The sets S_n, T_n ,

For Conjecture 1.3, take the coloring of probability measures $c(\nu) = \nu(Z)$ and any $0 < \varepsilon < 1/2$. Given every proposed increasing sequence of measures, the later proof recursively constructs admissible terms whose Z -values are arbitrarily close to zero and arbitrarily close to one. No single $r \in [0, 1]$ can put both within ε of r :

$$t(\mu_{i_k} \mid 1 \leq k < n)(T_p) > 1 - \epsilon.$$

Setting $i_0 := 0$, $s := \mathbf{1} * t$ and $(i_k \mid k < n)$ satisfy the conclusion of the claim. $\square$

The desired contradiction to Conjecture 1.3 of [5] now follows from Claims 1 and 3 with $m = 0$ by noting that for any $0 \leq r \leq 1$ and $0 < \epsilon < 1/2$, it is not possible for both 0 and 1 to be in the interval $[r - \epsilon, r + \epsilon]$.

Thus Conjectures 1.3 and 1.4 both have full negative literature answers.

4. SCOPE BOUNDARY

The source's concluding Questions 7.1–7.4 are broader than those two yes/no conjectures. The counterexample does not by itself decide every fixed number of summands, classify the binary systems for which an Ellis-type statement is true, or decide preservation under products. Nor does it settle the finite copy statement of Question 7.4: the source identifies that statement with amenability of Thompson's group F , and arXiv:1807.05469 explicitly says that the amenability question remains open. This packet therefore claims exactly the two central refutations and no more.

REFERENCES

- [1] J. T. Moore, *Hindman's Theorem, Ellis's Lemma, and Thompson's group F* , arXiv:1106.4735, Conjectures 1.3 and 1.4.
- [2] J. T. Moore, *Idempotent means on free binary systems do not exist*, arXiv:1807.05469, Sections 2–3.